Constraints:

1) every team plays with every other team exactly once
2) every team plays once a week
3) every team plays at most twice in the same period over the tournament.

Every problem solution is invariant through three different operations: the permutation of its rows, the permutation of its columns and the permutation of its team labels. All of these operations permute with each other, in a group theory sense, but not with themselves. Since rows and column permutations can be expressed via, respectively, right and left matrix multiplication with elemental swap matrices, this follows naturally from the associativity of matrix products. Then, since all permutations can be decomposed into compositions of transpositions, the fact that label permuting and row/column permuting commute can be proved manually for simple transpositions and it will ripple to every label permutation.
This makes it so that the group of invariance operations for a single solution is isomorphic to $$ S_n \times S_{n/2} \times S_{n-1}$$, a group with cardinality n!*(n/2)!*n-1!. This already large number would then have to be multiplied by 2^(n^2-n) if we hadn't split the solution and optimization components of problem solving (if we considered each match as an ordered pair rather than a two element set).
The goal of our symmetry breaking is, for all classes of matrices in n^(n^2-n) that are congruent under these operations, to elect a canonical representative. This will cut down our search space immensely, by only considering these instead of all possible matrices.

Classical:
n!, only need label fixing
label fixing + first column ordering, a bit less than n factorial

Miei:
2^(n/2)*(n/2)! di label fixing
FUCK THAT'S WHY

Given n-1 weeks, each period will see n-1 different matches, so 2n-2 team instances. Trivially, the constraint n.3 can only be satisfied with at maximum 2n team instances per period. So, out of this complete collection, only two team instances are missing. They can't be by the same team, because it would mean that that team would have to play n-1 matches in (n/2 - 1) periods, which without break either the constraint n.2 or n.3, since 2*(n/2 - 1) = n-2 < n-1 . This means that each period possesses exactly two teams, thus forth called defective, that only play once during it. Conversely, the constraint n.2 makes it so that each team has only one period it can be defective in.

The solution of the problem and the optimisation of its objective functions are two completely independent tasks, only connected by their shared parameter n. This is because we know a-priori that the schedule will contain all possible matches (as 2 elements subsets of the set of all teams) and the home or away state of a match does not interact with when it will occur (as it would have been, say, if we imposed that no team plays more than k matches in a row at home). This also means that, with a fixed n, once that an optimally balanced home/away match spread is obtained, it can be overlayed on any solution with no additional computation, making it optimal. Conversely, the construction of the problem solution has no bearing on optimality and all solutions have the chance to become optimal.